

Activity: Manage authorization

- Authorization is the concept of granting access to specific resources in a system. It's important because without authorization any user could access and modify all files belonging to other users or system files. This would certainly be a security risk.
 - In Linux, file and directory permissions are used to specify who has access to specific files and directories.
 - As a security analyst, setting appropriate access permissions is critical to protecting sensitive information and maintaining the overall security of a system.
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In this scenario, you must examine and manage the permissions on the files in the `/home/researcher2/projects` directory for the `researcher2` user.

The `researcher2` user is part of the `research_team` group.

You must check the permissions for all files in the directory, including any hidden files, to make sure that permissions align with the authorization that should be given. When it doesn't, you must change the permissions.

Here's how you'll do this task: **First**, you'll check the user and group permissions for all files in the `projects` directory. **Next**, you'll check whether any files have incorrect permissions and change the permissions as needed. **Finally**, you'll check the permissions of the `/home/researcher2/projects/drafts` directory and modify these permissions to remove any unauthorized access.

Task 1. Check file and directory details

In this task, you must explore the permissions of the `projects` directory and the files it contains. The lab starts with `/home/researcher2` as the current working directory. This is because you're changing permissions for files and directories belonging to the `researcher2` user.

1. Navigate to the `projects` directory.
2. List the contents and permissions of the `projects` directory.

```

researcher2@410876bd1a92:~$ cd projects
researcher2@410876bd1a92:~/projects$ ls
drafts project_k.txt project_m.txt project_r.txt project_t.txt
researcher2@410876bd1a92:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 26 07:43 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 26 07:43 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 26 07:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_t.txt
researcher2@410876bd1a92:~/projects$ ls -a
. .project_x.txt project_k.txt project_r.txt
.. drafts          project_m.txt project_t.txt
researcher2@410876bd1a92:~/projects$

```

ls -l is to show permission
ls -a is to show hidden files.
Hidden files start with (.)

3. Check whether any hidden files exist in the `projects` directory.

Which of these files is hidden in the projects directory?

- ☒ .project_x.txt
- ☐ .project_m.txt
- ☐ .project_r.txt
- ☐ There are no hidden files

Submit

What is the name of the group that owns the files in the projects directory?

- ☐ other_users
- ☐ security_team
- ☒ research_team
- ☐ researcher2

Submit

Task 2. Change file permissions

In this task, you must determine whether any files have incorrect permissions and then change the permissions as needed. This action will remove unauthorized access and strengthen security on the system. None of the files should allow the other users to write to files.

```
researcher2@410876bd1a92:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 26 07:43 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 26 07:43 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 26 07:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_t.txt
researcher2@410876bd1a92:~/projects$ chmod o-w project_k.txt
researcher2@410876bd1a92:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 26 07:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 26 07:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_t.txt
researcher2@410876bd1a92:~/projects$ chmod g-r , g-w project_m.txt
chmod: cannot access ',': No such file or directory
chmod: cannot access 'g-w': No such file or directory
researcher2@410876bd1a92:~/projects$ chmod g-r project_m.txt
researcher2@410876bd1a92:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 26 07:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 26 07:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_t.txt
researcher2@410876bd1a92:~/projects$
```

1. Check whether any files in the `projects` directory have write permissions for the owner of other.

Which file grants other users write permissions?

☐ project_t.txt

☒ project_k.txt

☐ project_m.txt

Submit

2. Change the permissions of the file identified in the previous step so that the owner type of other doesn't have write permissions.

```
chmod o-w project_k.txt
```



3. The file `project_m.txt` is a restricted file and should not be readable or writable by the group or other; only the user should have these permissions on this file. List the contents and permissions of the current directory and check if the group has read or write permissions.

What are the group permissions on the `project_m.txt` file?



☒ Read only

☐ Read, write, and execute

☐ Read and write

Submit

4. Use the `chmod` command to change permissions of the `project_m.txt` file so that the group doesn't have read or write permissions.

Click **Check my progress** to verify that you have completed this task correctly.



Change file permissions

Check my progress

You have completed this task and changed the permissions of files.

Task 3. Change file permissions on a hidden file

In this task, you must determine if a hidden file has incorrect permissions and then change the permissions as needed. This action will further remove unauthorized access and strengthen security on the system.

The file `.project_x.txt` is a hidden file that has been archived and should not be written to by anyone. (The user and group should still be able to read this file.)

```
researcher2@410876bd1a92:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 26 07:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 26 08:37 ..
-rw--w---- 1 researcher2 research_team  46 Jun 26 07:43 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 26 07:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 26 07:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_t.txt
researcher2@410876bd1a92:~/projects$ chmod g+r .project_x.txt
researcher2@410876bd1a92:~/projects$ chmod u-w .project_x.txt
researcher2@410876bd1a92:~/projects$ chmod g-w .project_x.txt
researcher2@410876bd1a92:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 26 07:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 26 08:37 ..
-r--r----- 1 researcher2 research_team  46 Jun 26 07:43 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 26 07:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 26 07:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 26 07:43 project_t.txt
researcher2@410876bd1a92:~/projects$
```

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2.1

1. Check the permissions of the hidden file `.project_x.txt` and answer the question that follows.

Which owner type has the incorrect write permissions?

- ☐ Just the group
- ☒ The user and the group
- ☐ Just the user

Submit

2. Change the permissions of the file `.project_x.txt` so that both the user and the group can read, but not write to, the file.

Note: Be sure to start the name of a hidden file with a period (`.`).

Click **Check my progress** to verify that you have completed this task correctly.



Change file permissions on a hidden file

Check my progress

You have completed this task and changed the permissions of hidden files.

Task 4. Change directory permissions

In this task, you must change the permissions of a directory. First, you'll check the group permissions of the `/home/researcher2/projects/drafts` directory and then modify the permissions as required. (You should be in the `projects` directory while managing the permissions of its subdirectory `drafts`.)

Only the `researcher2` user should be allowed to access the `drafts` directory and its contents. (This means that only `researcher2` should have execute privileges.)

```
researcher2@410876bd1a92:~/projects$ cd drafts
researcher2@410876bd1a92:~/projects/drafts$ ls -la
total 8
drwx--x--- 2 researcher2 research_team 4096 Jun 26 07:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 26 07:43 ..
researcher2@410876bd1a92:~/projects/drafts$ chmod g-x drafts
chmod: cannot access 'drafts': No such file or directory
researcher2@410876bd1a92:~/projects/drafts$ chmod g-x drafts.sh
chmod: cannot access 'drafts.sh': No such file or directory
researcher2@410876bd1a92:~/projects/drafts$ chmod g-x
chmod: missing operand after 'g-x'
Try 'chmod --help' for more information.
researcher2@410876bd1a92:~/projects/drafts$ chmod g-x drafts
chmod: cannot access 'drafts': No such file or directory
researcher2@410876bd1a92:~/projects/drafts$ cd projects
-bash: cd: projects: No such file or directory
researcher2@410876bd1a92:~/projects/drafts$ cd
researcher2@410876bd1a92:~$ cd projects
researcher2@410876bd1a92:~/projects$ chmod g-x drafts
researcher2@410876bd1a92:~/projects$ cd drafts
researcher2@410876bd1a92:~/projects/drafts$ ls -la
total 8
drwx----- 2 researcher2 research_team 4096 Jun 26 07:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 26 07:43 ..
researcher2@410876bd1a92:~/projects/drafts$
```

X is
permission
to execute

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1. Check the permissions of the `drafts` directory and answer the following question.



Does the group have permissions set to access the drafts directory and its contents?

☒ Yes

☐ No

Submit

2. Remove the execute permission for the group from the `drafts` directory.

Click **Check my progress** to verify that you have completed this task correctly.



Change directory permissions

Check my progress

You have completed this task and changed the permissions on the drafts directory.

Current file permissions

This document displays the file structure of the `/home/researcher2/projects` directory and the permissions of the files and subdirectory it contains.

In the `/home/researcher2/projects` directory, there are five files with the following names and permissions:

- `project_k.txt`
 - User = read, write,
 - Group = read, write
 - Other = read, write
- `project_m.txt`
 - User = read, write
 - Group = read
 - Other = none
- `project_r.txt`
 - User= read, write
 - Group = read, write
 - Other = read
- `project_t.txt`
 - User = read, write
 - Group = read, write
 - Other = read
- `.project_x.txt`
 - User = read, write
 - Group = write
 - Other = none

There is also one subdirectory inside the `projects` directory named `drafts`. The permissions on `drafts` are:

- User = read, write, execute
- Group = execute
- Other = none